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EXTREMELY CONFIDENTIAL

Palm Harbor, Florida, January 20, 2004

THE NEW HADRONIC POWER PLANTS

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President

The Institute for Basic Research

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SUMMARY

Available power plants essentially burn coal producing steam that powers a turbine for the generation of electricity or other uses. Since these plants are no longer environmentally acceptable due to the release into our atmosphere of excessive toxic substances, depletion of breathable oxygen and other serious environmental problems, the author has developed over the past two decades of research new means of power generation called "Hadronic Power Plants" (patented in U.S.A., international patents pending) based on the combustion of coal via electric arcs submerged in a properly selected liquid or liquid waste.

Hadronic Power Plants are environmentally acceptable because they produce two clean sources of energy: a large amount of heat, and the clean burning MagneGas(TM) fuel. Self-sufficiency in the production of the electricity used by the arc is reached via the use of heat and part of the MagneGas produced. The remaining MagneGas is used for the production of green electricity for the grid or other uses. The only remaining byproduct is given by carbonaceous solid precipitates that are re-used for the production of electrodes.

In this way, Hadronic Power Plants produce no solid, liquid, gaseous, noise or other form of pollution. Moreover, the combustion of coal by the arc occurs via oxygen originating from liquids, rather than from our atmosphere as in conventional coal powered plants. As a result, Hadronic Power Plants allow a significant reduction of the alarming oxygen depletion already caused by fossil fueled plants.

The understanding of the new Hadronic Power Plants requires the admission that the main reactions and energy balance in a conventional power plant burning coal via flames in a furnace and the corresponding chemical reactions and energy balance of the new Hadronic Power Plants burning coal via an electric arc are essentially the same. Therefore, any doubt about the capability of producing energy via the arc combustion of coal is a denial of the evidence that the combustion of coal produces a large amount of energy.

The new power plants are called "hadronic" because quantum mechanics, quantum superconductivity and quantum chemistry imply excessive deviations from experimental data of submerged electric arcs reviewed below, thus requiring for their quantitative or otherwise credible treatment the covering "hadronic mechanics," "hadronic superconductivity" and "hadronic chemistry." The latter new disciplines were developed during the past two decades thanks to the contribution by a large number of mathematicians, theoreticians and experimentalists from all developed countries precisely for the quantitative study of "new" energies, that is, energies that cannot be predicted or treated via old quantum disciplines.

In this very confidential presentation the author reviews the basic principles of Hadronic Power Plants, identifies the status of the development in the field, suggests specific further tests, and provides a projection of their cost.

SECTION 1: INTRODUCTION

1.1: Environmental Problems Caused by Fossil Fuels.

Due to the general lack of detailed information on environmental issues, the understanding of the need for the new Hadronic Power Plants requires a detailed knowledge of the real environmental problems caused by fossil fuel combustion.

According to official data of the U. S. Department of Energy, we consume today in our planet about 4 trillion gallons of gasoline per day (or about fifteen trillion liters of gasoline per day), plus an essentially similar consumption of coal and natural gas. This disproportionate consumption of fossil fuels via the traditional processes of the past century has caused serious environmental problems only some of which are disclosed to the general public, including:

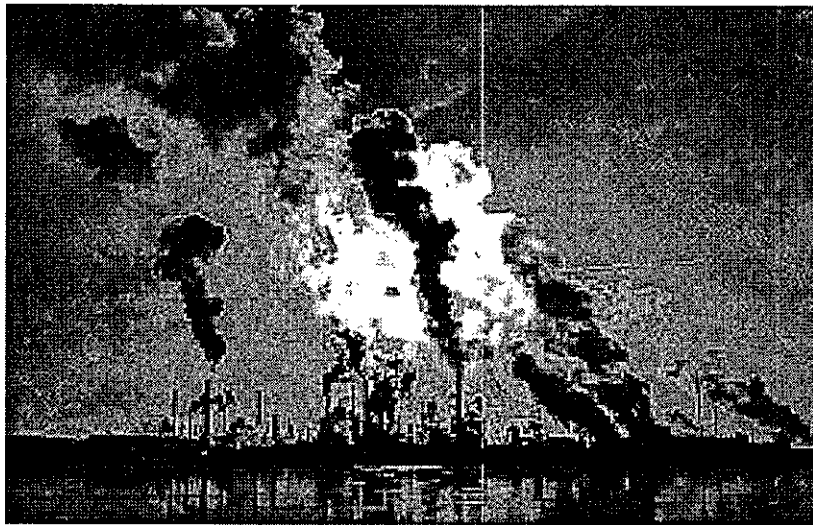


FIGURE 1: A view of a coal fueled electric power plant

1) **The green house effect caused by an alarming increase of CO₂ in our atmosphere.** This first environmental problem is known to the general public, although without technical details. The percentage of CO₂ in our atmosphere at sea level in 1950 was 0.033 % $\pm$ 0.01 % (see, e.g., Encyclopedia Britannica of that period). According to various measurements, the current CO₂ percentage in our atmosphere is 50-100 times that of 1950. Simple chemical calculations presented in detail on the web site [1] establish that we today release into the atmosphere about 150 thousands metric tons of CO₂ per day, of which only 80,000 tons are estimated to be recycled by plants into breathable oxygen by the chlorophyll process, while the remaining 70,000 tons of CO₂ per day remain untreated by nature, thus causing the green house effect. Besides for lakes in the North Pole created by melted ice during summer, 500 deaths in the Southern U.S.A. caused by tornados in the winter of 2003, some 15,000 deaths in the summer of 2003 in Europe

caused by excessive heat, simple computer simulations predict that certain regions will become un-inhabitable in certain periods of the year because of cataclysmic climactic effects. CO₂ detectors are now available in battery operated portable versions of low cost. To avoid the rather widespread political manipulation of environmental issues, individuals seriously committed to the environment are suggesting purchasing such a detector and personally conducting local measurements of CO₂ excess in our atmosphere.

2) The permanent removal of breathable oxygen from our atmosphere, called by the author the “oxygen depletion” [1]. The general public has been kept unaware of the fact that fossil fuels require oxygen to burn, thus removing breathable oxygen from our planet. In fact, the “O₂” in the excess “CO₂” now existing in our atmosphere was originally breathable oxygen. The oxygen percentage in our atmosphere in 1950 was 20.946%. No measurement of the oxygen depletion caused by fossil fuels has been officially released by governments from currently available satellite and other detections. We only have rumors of a global depletion of 3% to 5% breathable oxygen in our atmosphere. However, simple chemical calculations from the CO₂ excess also done in [1] show that today we remove from our atmosphere an estimated 50 million metric tons of breathable oxygen per day (the percentage of O₂ in the non-recycled CO₂ excess). Since the total volume of oxygen in our atmosphere is known, simple predictions indicate that, at the currently increasing combustion of fossil fuels, oxygen depletion will cause a proliferation of heart problems within one decade, the need for oxygen bottles for any human activity within three decades, and the eventual extinction of the human race within one century. Most people, including most environmentalists, often forget that we need oxygen to live. O₂ detectors are also available in battery operated portable versions at low cost. Individuals seriously committed to the environment, as well as to their own health, are suggested to purchase such a detector and personally conduct local measurements of O₂ deficiency in our atmosphere.

3) Excessive release of carcinogenic substances in the combustion exhaust. Other information carefully suppressed to the general public is that fossil fuel exhaust is the biggest source of cancer on Earth today, since said exhaust is highly carcinogenic and is inhaled by all of us every day in amounts that are at least 10,000 times bigger than toxic substances ingested in food. The name “atmospheric pollution,” “hydrocarbons” and other names used for the combustion exhaust of fossil fuels are offensive, since they are synonyms for highly carcinogenic and toxic substances. Simple calculations done in [1] establish that we release in our atmosphere about 30 million metric tons of carcinogenic and toxic substances per day. These substances, being mostly stable gases, constitute an increasing permanent health problem for us, let alone for our children and grandchildren. Individuals seriously committed to the environment are suggested to secure chemical analyses of fossil fuel combustion exhaust available on various web sites, and submit them to their medical doctors for identification of their carcinogenic content. For example, the origin of the recent large proliferations of breast cancer has been identified not only in gasoline exhaust, but actually in breast cells that are particularly receptive to specific components of gasoline combustion exhaust.

As an additional part of the widespread misinformation existing about environmental issues, it is generally believed that the combustion of **natural gas** alleviates the above environmental problems. With the understanding that the combustion of natural gas is better than that of coal or heavy oils, this belief is generally false. It is true that the "flame" of natural gas is cleaner than that of gasoline. However, gasoline is a liquid while natural gas is a gas. Therefore, gasoline is about 1,500 times denser than natural gas. When natural gas and gasoline are compared via "measurements" (rather than pep talks) for the "same performance under the same conditions" (rather than via political comparison of flames), natural gas results in being much more polluting than gasoline because it implies much bigger releases of CO₂ and toxic substances and a bigger oxygen depletion (see the comparative "measurements" of natural gas-gasoline done by an EPA accredited laboratory in Ref. [2]).

The climax of misinformation to the public is reached with **hydrogen** because it is claimed to resolve all environmental problems caused by fossil fuel combustion on grounds that the hydrogen combustion exhaust is solely composed of water vapor H₂O. It is true that hydrogen is environmentally acceptable under the following conditions: 1) Hydrogen is produced via the electrolytic separation of water; 2) Oxygen produced in this way is freely released in the atmosphere to avoid oxygen depletion; and 3) The used electricity is of clean renewable type such as wind, solar or hydric nature.

However, in reality the use of hydrogen causes very alarming oxygen depletion because all oxygen produced via electrolytic processes is sold and certainly not freely released in the atmosphere, while no governmental regulation of this aspect so vital for humanity is even under consideration at this time by any government on Earth. Under these conditions, the combustion of hydrogen implies very alarming oxygen depletion because the "O" in the water vapor "H₂O" of hydrogen combustion exhaust was originally breathable oxygen permanently turned into water. By comparison, gasoline combustion produces CO₂ that is at least partially recycled by plants into breathable oxygen, for which reason we are still alive today despite the disproportional combustion of fossil fuels. It is known by technicians that, in the event all cars on Earth were fueled by hydrogen as currently produced, life will be terminated because of lack of oxygen to breathe.

In the absence of new clean power plants, large scale production of hydrogen will mandate the use of fossil fueled electric plants. Since electrolysis and other hydrogen production processes are highly inefficient, the environmental problems 1), 2) and 3) above caused by the production of hydrogen will be bigger than the direct combustion of gasoline (for more details, see the home page of [3]).

Other solutions currently under consideration for the solution of the above environmental problems must be dismissed for various reasons. For instance:

A) After spending hundreds of billions of dollars in public funds, the "**hot fusion**" has been proved to have no industrial value because of the excessive temperatures causing uncontrollable instabilities at the initiation of the fusions themselves;

B) After spending additional large sums (this time, of essentially private origin), the **"cold fusion"** has proved the capability to synthesize nuclei in our laboratories on Earth. However, the cold fusion must also be dismissed as an industrially viable source of clean energy because of the excessively low energy gains. The reason is essentially opposite to the failure of the hot fusion, namely, the insufficiency of the temperature to expose nuclei, in which case no systematic nuclear fusion is possible because it is prohibited by the atomic electron clouds shielding the nuclei.

C) Other attempts, such as those seeking to utilize the **"zero-point energy"** are faced with very serious theoretical, yet unsolved problems (such as the general violation of the principle of conservation of energy), to prevent serious industrial consideration at this time. It is true that empty space is the biggest reservoir of energy known to mankind. However, we know at this moment of no mechanism capable of extracting energy from empty space. Until such a mechanism is first identified, efforts based on using unreachable energies cannot be part of serious planning for the solution of serious environmental problems.

The above extremely un-reassuring scenario establishes beyond any possible doubt the **need to develop new clean sources of power**. In this respect we should denounce the rather general posture of sitting and waiting for others to resolve the problems. Environmental problems invest all of society. Their solution can only result from a collegial effort based on the failure of all existing attempts outlined above; it is also easy to predict that no solution is conceivably possible without adequate funding of qualified research on basically NEW processes and methods.

1.2: Coal Combustion via Submerged Electric Arcs

After searching for decades, new clean sources of power are industrially relevant now with available technologies, the author selected the combustion of carbon via electric arcs submerged within oxygen rich liquids, such as water or water-base liquid wastes. The selection was based on a number of reasons that can be outlined as follows:

The energy produced by the combustion of coal in a furnace essentially originates from the combination of carbon and atmospheric oxygen into CO. This chemical reaction is exoenergetic, by releasing 255 Kcal/mole. In turn, CO is combustible and produces CO₂ (a natural byproduct of fossil fuel combustion jointly with CO) with an additional 84 Kcal/mole. The release of carcinogenic and other toxic substances are due to the poor and incomplete combustion of coal (in fact, the HC are precisely unburned debris of coal). The oxygen depletion is caused by the fact that the oxygen needed for the combustion originates from our atmosphere.

By comparison, submerged electric arcs decompose the water molecule in its atomic constituents, vaporize and ionize the same, while jointly vaporizing and ionizing the carbon electrodes, by forming plasma at about 10,000 degrees F of mostly ionized atoms of H, C and O. In a stationary arc, C and O combine exactly as for the combustion of coal, by equally releasing 255 Kcal/mole. Moreover, the presence of CO within the

electric arc and oxygen produces CO₂ with the additional release of 84 Kcal/mole also as for the combustion of coal in a furnace. In addition, H and H recombine to produce H₂ with the release of 105 Kcal/mole not occurring in the combustion of coal in a furnace. Finally, H and O partially recombine into H₂O with the release of additional 57 Kcal/mole that are also absent in the combustion of coal in a furnace. In summary, the chemical reactions in the plasma of carbon electrodes produce 112 Kcal/mole MORE than the combustion of coal in a furnace.

The resulting products are the following: firstly, there is the production of a large amount of heat acquired by the liquid that can be used via heat exchangers; secondly, there is the production of a clean burning combustible gas that bubbles to the surface of the liquid for collection and use; thirdly, there is the release of carbonaceous precipitates that can be collected via filtering of the liquid and used for the production of electrodes.

The first point important for this presentation is that the overall energy balance by the combustion of coal with an electric arc is bigger than that of the combustion of the same coal in a furnace. This is due to the fact that the energy released by the submerged electric arc is a multiple of the electric energy used by the arc itself. In turn, this occurrence is due to the fact that in an open arc submerged in water the electric resistance is very high and of the order of 100 Ohms (evidently because pure water is an excellent electric insulator). As soon as the arc is closed and becomes stationary, the resistance drops down to fractional Ohms depending on water impurities and other factors. This establishes that water under a DC electric arc becomes the best known (but generally ignored) superconductor, since it requires minimal electric energy for its maintenance, as documented below.

Additional reasons for the bigger energy release are due to the very inefficient nature of the combustion of coal in a furnace compared to the arc combustion. In fact, the very polluting character of conventional coal combustions is a direct measure of the inefficiency of the combustion itself. By comparison, arc combustion of coal is more efficient and, consequently, releases more energy. Also, conventional coal combustions release large percentages of coal debris in the exhaust, while arc combustion of coal releases no carbon in the atmosphere, since carbonaceous precipitates in the liquid are filtered and used for electrode production.

Moreover, submerged electric arcs between coal electrodes have the following environmental advantages: the combustion of coal does not cause oxygen depletion because oxygen originates from liquids and not from our atmosphere; said combustion does not produce any carcinogenic or other toxic substances at all, because the produced combustible gas cannot possibly contain any HC since it is formed at the 10,000 degrees F of the electric arc.

The above occurrences establish the main principle of this presentation, namely, that the combustion of coal via an electric arc can indeed generate power in addition to that needed for its own operation. The only debatable issue is the selection of the appropriate

technological realization of a power plant operating with submerged electric arc, and the availability of the necessary funds for its development.

1.3: Coal Combustion via the PlasmaArcFlow(TM) Process

Submerged electric arcs as well as the combustible character of the gas they produce have been known for about 150 years. In fact, sailors discovered in the middle of the 19-th century that the gas bubbling to the surface from undersea welding is combustible. Despite that, no industrial or environmental maturity was reached by the conventional submerged arc technology for the following reasons.

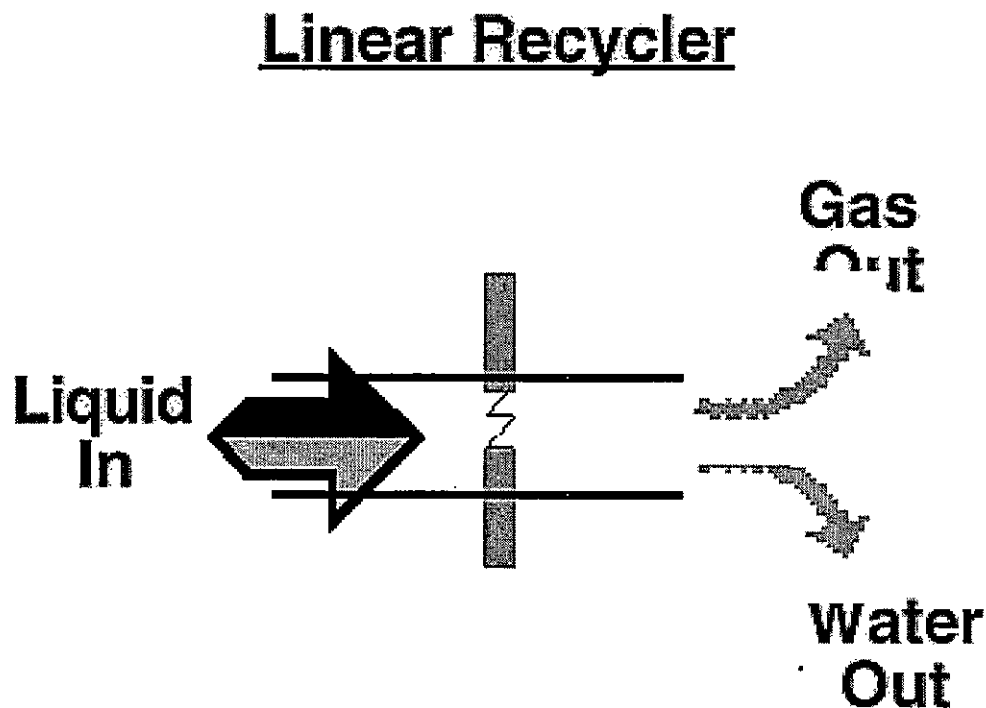


FIGURE: A schematic view of the PlasmaArcFlow Process

While the electric arc is, by far, the most effective known process to separate water, a stationary electric arc has a very low efficiency because most of the H and O obtained from water separation recombine into H₂O via an implosion, which recombination is the origin of the beautiful glow of underwater electric arcs. Various measurements indicate that the energy produced via a conventional submerged electric arc is in effect marginal.

The process is also not acceptable from an environmental profile because, even though the process implies an important reduction of oxygen depletion and the elimination of toxic releases in the atmosphere, a stationary electric arc produces a gas with up to 15% CO₂ content “before” its combustion, with up to 23% CO₂ in the combustion exhaust. Since CO₂ is the origin of the green house effect, such a high CO₂ content renders the

gas environmentally unacceptable. In view of these and other insufficiencies, the author developed a "new process" based on submerged electric arc, known today as "Santilli PlasmaArcFlow(TM) Process" (patented in the U.S.A., international patents pending), and based on the flowing of the liquid feedstock through the electric arc by therefore continuously removing the plasma from the tip of the electrodes immediately following its creation. The gas produced with the PlasmaArcFlow Process is chemically different from that produced by a stationary arc, and has been called "MagneGas(TM)" because of certain magnetic polarizations outlined on the web site [3].

This new PlasmaArcFlow Process significantly reduces the recombination of H and O into H₂O, with consequential increase of the energy efficiency. At the same time, the PlasmaArcFlow Process prevents the CO from burning into CO₂ by essentially eliminating the CO₂ content in the gas itself.

Other advantages of the PlasmaArcFlow Process are due to the lack of creation of the conventional CO (the one with triple valence bonds), in favor of single bonded C-Ö and double bonded C=O. The latter are notoriously unstable and decompose during combustion, rather than producing CO₂, thus permitting the release of a large percentage of breathable oxygen in the exhaust, while decreasing its CO₂ content.

Various measurements have established the increase of energy efficiency by a significant numerical factor in the transition from a stationary submerged electric arc as the same treated with the PlasmaArcFlow Process. Similarly, the presence of CO₂ in the gas produced is a direct measure of the insufficient flow. For these and other aspects, we refer the reader to the web site [3].

It should be indicated that the above outline refers to PlasmaArcFlow processes based on a DC electric arc. Significant increases of the efficiency are possible for other arc processes that cannot be disclosed at this time because it is being developed.

1.4: The Three Types of Hadronic Power Plants

MagneGas fuel does indeed provide a solution to the serious environmental problems outlined in Section 1.1. In fact, MagneGas combustion exhaust is composed of 50% water vapor; 14%-18% breathable oxygen; 5%-7% CO₂; the rest is given by atmospheric gases without toxic substances. Note that the small percentage of CO₂ is tolerable and, because it is small, it can also be removed via CO₂ absorbing sponges in the exhaust. Also, MagneGas is cost competitive with respect to fossil fuels if produced in sufficient volume, and can be used by any existing internal combustion engine.

By far the most profitable part of the MagneGas Technology is the production of hydrogen via its separation from MagneGas. In fact, industrial-grade hydrogen sells today in the USA at about 50 times the cost of natural gas energy equivalent (H₂ sells at \$0.15/scf with 300 BTU/scf while C-H₄ sells at \$0.01/scf with 1,000 BTU/scf), with bigger selling prices occurring for pure hydrogen as needed for fuel cells. Since hydrogen is contained in MagneGas in a mixture that can be easily separated via molecular sieving

processes with minimal cost, the MagneGas Technology is the only one known today permitting a dramatic reduction of hydrogen production costs with large profits.

Also, PlasmaArcFlow Reactors are at least 10 times more efficient than electrolysis and other hydrogen production methods, thus implying a major reduction of the pollution caused by hydrogen production. Finally, the MagneGas Technology is based on the principle of using it for combustion oxygen of liquid (rather than of atmospheric) origin. In fact, the combustion exhaust of the gas remaining from the separation of hydrogen from MagneGas is extremely rich in oxygen. Therefore, the production of hydrogen from MagneGas does indeed alleviate the biggest environmental problem of the emerging new hydrogen era, oxygen depletion.

The MagneGas Technology has now reached industrial maturity for large profits via the recycling of liquid waste and the production and sale of MagneGas, hydrogen and other specialty fuels, or the sale of the green electricity they produce.

Despite the large efficiency of PlasmaArcFlow Reactors that operate at a fraction of the electric energy used by other fuel technologies, such as electrolysis, the MagneGas Technology remains dependent on the use of electricity. Therefore, a primary societal need is the development of new clean source of electricity.

This important societal need can indeed be fulfilled by the MagneGas Technology due to the large efficiency of its PlasmaArcFlow Recyclers that, depending on proper funding to complete the development, can indeed produce all the electricity needed to operate their arcs, and produce additional MagneGas for the clean production of electricity to be released in the grid. As a matter of fact, this new source of electricity can be achieved via three progressive reactors as outlined below.

The HADRONIC POWER PLANTS OF FIRST TYPE are here defined as those based on the use of electric arcs for the combustion of coal electrodes via the sole use of the PlasmaArcFlow Process operated at suitable power, pressures and temperatures. As discussed in Sections 2 and 3, this first type of Hadronic Power Plant is indeed capable of self-sufficient production of power via a DC electric arc, with increased power output for the above indicated new arc processes currently under development.

The fuel produced by a submerged electric arc is at the gaseous state. A general chemical law establishes that all combustible gases can be turned into a liquid form via a catalytic (rather than a cryogenic) process occurring via the exposure of the gas to a number of catalysts (varying from gas to gas), the exposure occurring at pressure generally of 250 psi or 15 bars. The process was discovered by Germans during WWII and it is done via Fisher-Trops Catalytic Towers.

Basic tests have established that MagneGas can also be liquefied catalytically, the sole issue open for industrial liquefaction being the selection of the best catalysts for optimal liquefaction. The implied transition of state from gas to liquid "releases" large amounts of heat that can be added to the heat produced by the electric arc for the completely self-

sufficient generation of power, as described in detail in the third patent of the MagneGas Technology, Ref. [4].

It should be noted that PlasmaArcFlow can well operate at the pressures needed by Fisher-Trophy towers without any compression. Therefore, the catalytic liquefaction towers of MagneGas merely are additional stations that can be added to or integrated with the reactors.

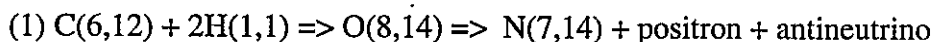
The HADRONIC POWER PLANTS OF THE SECOND TYPE are here defined as those based on the self-generation of the electricity used by the arc via the heat produced by the PlasmaArcFlow Process, plus the heat produced by the catalytic liquefaction of MagneGas, while the resulting liquid fuel, called "MagneLiquidTM," is used for the production of usable electricity or other applications.

Note that the U. S. Patent Office did indeed grant a patent on the self-sufficient generation of power because of the large amount of heat produced by the arc and the catalytic liquefaction.

Despite the achievement of self-sufficiency in the generation of clean power with the above two processes, the true reasons for the selection by the author of the submerged electric arc, here disclosed for the first time, are the following:

Chemical analyses of air bubbles trapped in amber have established that about 100 million years ago our atmosphere had only 40% of nitrogen. Similar measurements have also established that the increase of nitrogen in our atmosphere has been gradual, thus excluding a collision of Earth with a nitrogen rich comet or astrophysical body, assuming that such a body exists. Similarly, volcanic eruptions have been excluded because it is not emitting nitrogen in such a quantity as to explain the doubling of the percentage of nitrogen in our planet in about 100 million years.

The above occurrence establishes the existence in our atmosphere of a process that slowly synthesizes nitrogen. After considerable study, the author has proved that nitrogen is synthesized by lightning from carbon and hydrogen according to reaction

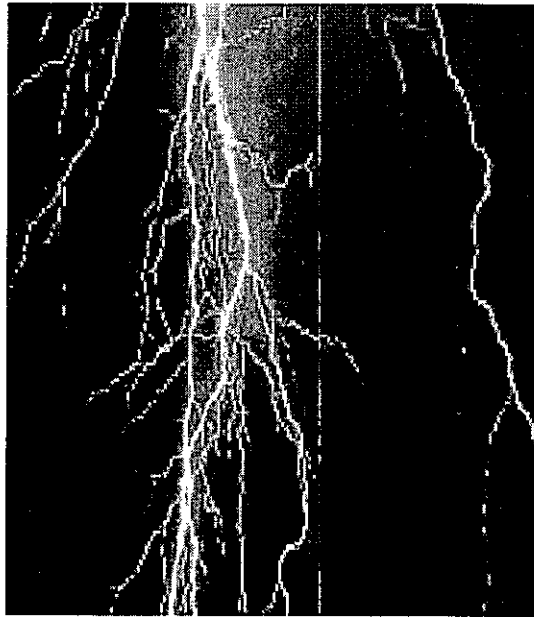


where $\text{O}(8,14)$ is notoriously very unstable and decays in the indicated mode (see nuclear tables). Other syntheses are also possible, but are here ignored for simplicity. Therefore, nitrogen is synthesized any time lightning hits a carbon-base substance such as vegetation, since hydrogen is automatically produced by lightning in the separation of atmospheric vapor and is available everywhere along the discharge.

Independently from the above, it is today known that the extreme amount of energy needed for a scientific (that is, numerical) representation of thunder simply cannot be explained via conventional chemical reactions caused by lightning, because of the very small sectional area of lightning combined with its extremely short duration.

Studies conducted by the author have reached the first known numerical representation of thunder as being caused by the very large energies released by nuclear syntheses such as (1). Simple calculations show that the synthesis of nitrogen with the efficiency of $1/10,000,000,000,000$ atoms is sufficient to generate 1,000,000 Joules of energy. This origin of thunder is confirmed also by other factors, such as the fact that thunder becomes smaller when lightning occurs in between clouds without contact with carbon rich substances, in which case nuclear synthesis (1) does not occur, and we have other syntheses with smaller probability.

The author selected the submerged electric arc for a new generation of clean power plants because the submerged electric arc is the best form of "controlled lightning," as well as for the fact that the efficiency in nuclear syntheses is proportional to the density of the medium in which the arc occurs. In this way, the increase of the density of the medium in which the submerged electric arc occurs can compensate for the inability to reach arcs with millions of volts.



FIGUREA: A view of lightning.

The HADRONIC POWER PLANTS OF THIRD TYPE are here defined as those in which the primary source of energy is given by the nuclear syntheses permitted by the arc, the combustible gas produced being a secondary source. The reader is certainly aware of the fact that this third type of Hadronic Power Plants, if properly developed, can lead to virtually unlimited amounts of clean energy because, as indicated in Section 2, systematic studies of the nuclear syntheses permitted by submerged arcs have established other syntheses besides (1) being mostly dependent on the selected liquid feedstock and other factors.

Due to the "fission" of large and heavy nuclei, nuclear power plants are not environmentally acceptable because of the release of highly radioactive waste, as well as dangerous radiations. By comparison, since they are based on the "synthesis" of light,

Immediately following the detection of nuclear syntheses, the author ordered extensive measurements of possible radiations released by the PlasmaArcFlow Reactors that were conducted by the Florida Radiation Association. Their certification of complete absence of radiation reproduced in [3]. At any rate, no radiation is possible by synthesis (1) since the sole radiation is given by electrons and photons that are trapped by the liquid itself, besides from the innocuous neutrinos (assuming that they exist - see [7] for technical reasons of doubt).

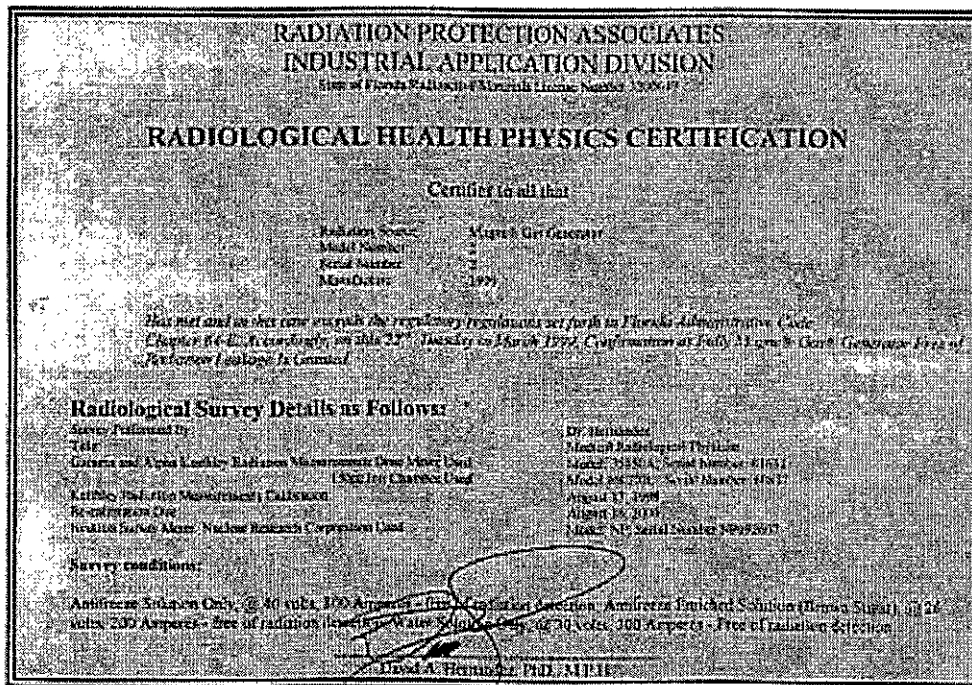


FIGURE: A view of the certification that PlasmaArcFlow Recyclers Release no radiation of any type.

The topic of this presentation cannot be disclosed to academia because experimental evidence on submerged electric arcs disproves quantum theories to such a degree that it cannot be accommodated via the usual addition of unknown parameter and similar manipulation to fix deviations. The huge financial, academic and other organized interests in quantum theories request the absolute silence about the topic of this presentation to academia due to its easily predictable unethical use of any means to discredit, jeopardize or disrupt the research irrespective of its societal values. Some of the deviations can be summarized as follows:

- 1) As it is well known, quantum mechanics and chemistry predict that the combustible gas produced by an electric arc between carbon-base electrodes submerged in ordinary

water is composed of 50% (in volume) H_2 and 50% CO. But CO is combustible and produces CO_2 . Therefore, in the event the above belief were true, the combustion exhaust of said gas should contain about 40% CO_2 . Measurements have established that the CO_2 content of the combustion exhaust can be as low as 5%, thus establishing a ten-fold error under which any use of quantum mechanics and chemistry is a scientific fraud.

2) As recalled above, the formation of CO is extremely exothermic, by releasing 255 Kcal/mole. Calculations done by Russian chemists, and reported in Section 7.10 of monograph [5], have established the prediction by quantum mechanics and chemistry that the production of one standard cubic foot (scf) of MagneGas should be joined with the production of 2,250 British Thermal Units (BTU) of heat acquired by the liquid when operated at atmospheric pressure. Actual measurements have instead established that we have the production of about 250 BTU/scf, thus implying a ten-fold error, under which the use of quantum mechanics and chemistry constitutes a scientific fraud. Note that the latter error is indirect, thus being fully compatible with, and actually confirming the preceding discrepancy that is in excess. In fact, MagneGas contains about 5% of CO and not 50%.

3) Quantum mechanics and chemistry predict that the combustion exhaust of a gas produced from a submerged arc within pure water is solely composed of water vapor (from the combustion of hydrogen), CO_2 (from the combustion of CO) and the rest is given by atmospheric gases without appreciable oxygen. Measurements establish instead that the combustion exhaust of MagneGas contains up to 18% of breathable oxygen, thus implying an error of such a magnitude under which the use of quantum mechanics and chemistry constitutes a scientific fraud.

4) Quantum mechanics and chemistry predict that the hydrogen contained in the combustible gas produced from an underwater electric arc has the conventional molecular structure with a valence bond $H_2 = H-H$, with standard specific weight of 2.016 atomic mass units (amu). Measurements have instead established that the hydrogen separated from said gas has a multiple of the specific weight of H_2 [6], under which conditions any use of quantum mechanics and chemistry is a scientific fraud. In fact, this evidence permitted the discovery of the first new chemical species not based on valence called "MagneCules" since the identification of molecules by Avogadro, Cannizzaro and others in the 18-th century.

5) Quantum mechanics and chemistry absolutely prohibit nuclear syntheses of type (1) caused by lightning (because they are only admitted under the extreme pressures and temperature in the core of stars). Measurements have established the existence of nitrogen in MagneGas in a large multiple of the very small percentage of atmospheric nitrogen occluded in liquids, under which conditions any use of quantum mechanics and chemistry for the study of nuclear syntheses of type (1) constitutes a scientific fraud.

The resolution of the above large deviations is the origin of the construction of hadronic mechanics, hadronic superconductivity and hadronic chemistry (see monograph [5] for a post Ph. D. level presentation and web site [7] for a non-technical review).

Scientists, business people and politicians interested in the environment are discouraged from using or suggesting the use of quantum theories in the study of submerged electric arc technologies to avoid a clear waste of money, let alone a proliferation of misinformation. At any rate the author wants to have no relationship with such studies due to the implied unethical misconduct.

The impossibility of using quantum theories for submerged arc technologies and the need for suitable covering theories is beyond credible or otherwise scientific doubt. The only debatable issue is the selection of the appropriate generalization of quantum theories, but not the need for such a generalization.

The author uses hadronic theories for the study of submerged electric arcs because they are the ONLY ones available at this writing that have been verified by a large number of experimental evidence in various fields, they are the ONLY ones that have resolved inconsistencies 1) to 5) above; and they are the ONLY ones possessing the same invariance of the numerical predictions as quantum theories [5, 7].

At any rate, the basic equations of hadronic theories have been proved to be "directly universal," in the sense of including as particular cases "all" possible generalizations of quantum equations ("universality"), directly in the coordinates of the experimenter without any need for transformation theory ("direct universality"). As a result, the author does not intend to be part of a study of other generalizations of quantum theories.

Note that in the material released to the public [1, 2, 3] there is no indication of the intrinsic self-sufficiency of the PlasmaArcFlow Process. This is due to the decay of the scientific community in which new advances are generally dismissed as fraud without any prior inspection or verification.

In view of this well known ethical decay, the U. S. military were forced in the 1970s to terminate the funding of all academic research and conduct research in their own laboratories. Today, the research conducted at military laboratories, such as Sandia Labs, is immensely more advanced than the research conducted at MIT, Harvard, Princeton, Berkeley and similar places around the world. In fact, academia has produced absolutely no truly "new" discovery in the last 50 years because of the totalitarian control of the research by pre-existing pet theories, while the discoveries by military laboratories are due to the lack of such political control with consequential results simply beyond our imagination.

Following the military example, U.S. corporations now avoid any disclosure of basic advances to academia, as an evident necessary condition to avoid damage caused by irresponsible behavior by vociferous academicians in a posture of scientific power and credibility. Academia anywhere in the world must be kept outside any participation in research outlined in this presentation, as a necessary condition for its very conduction and survival. The reader should be made aware that, in the event of any disclosure of the

content of this presentation to any member of academia, the author shall terminate any further collaboration.

SECTION 2: HADRONIC POWER PLANTS

2.1: Introduction

The scope of this presentation is to illustrate the existence of a hierarchy of new, clean, self-sufficient power plants, to propose further developments, and to identify the related costs.

Within such an objective, it is important to identify first the self-sufficiency of Hadronic Power Plants of the First Type, those without any catalytic liquefaction or nuclear syntheses. Once the self-sufficiency is established at that level, it can only improve at the subsequent levels. In turn, the understanding of the efficiency of said first type requires knowledge of the efficiency of the underlying reactors and how it can be maximized.

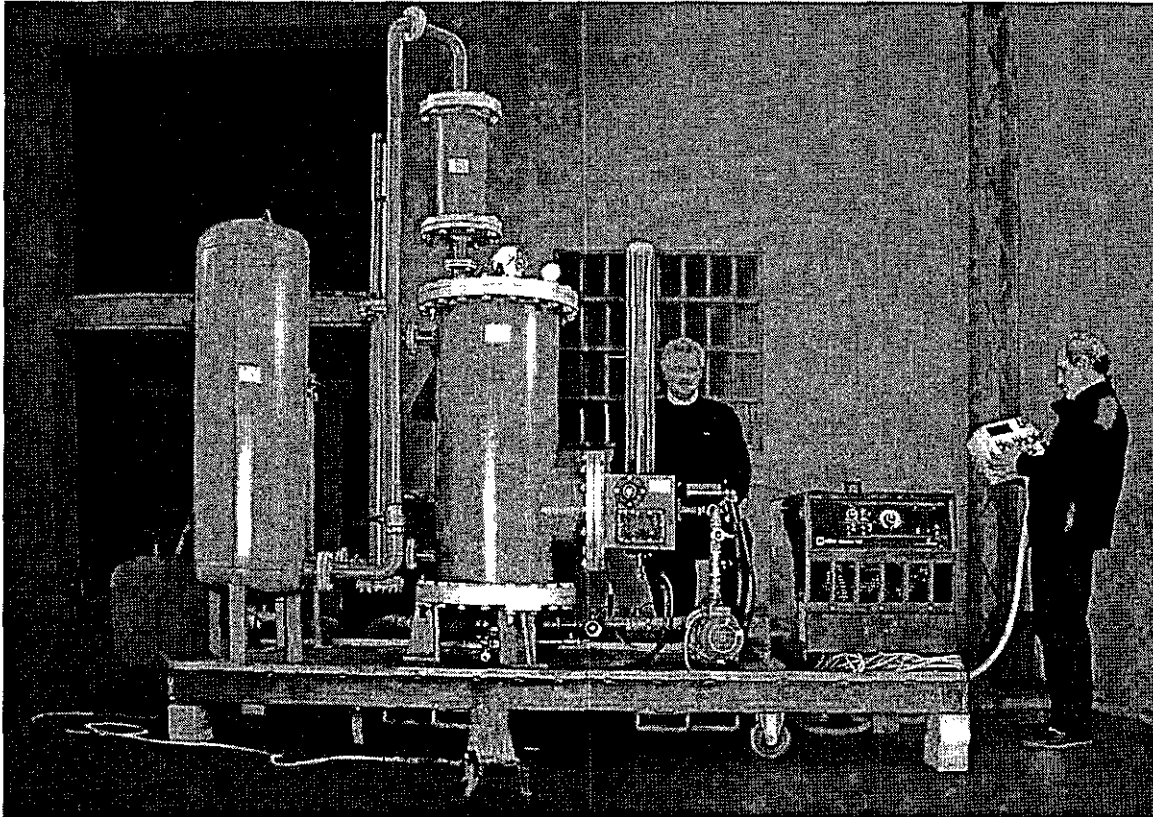


FIGURE: A view of the 50 KWH Verona Reactor

Therefore, we shall show that the capability of self-sufficient production of clean power already exists for the Hadronic Power Plants of the First Type, and can be established via measurements with the existing PlasmaArcFlow Recyclers, such as the 50 KWH Recycler available in Verona Italy, the 100 KWH industrial recycler available in Florida, or the research recycler available in Kazakhstan, provided that sufficient funds are available for the purchase of the necessary meters and the payment of the related operating costs.

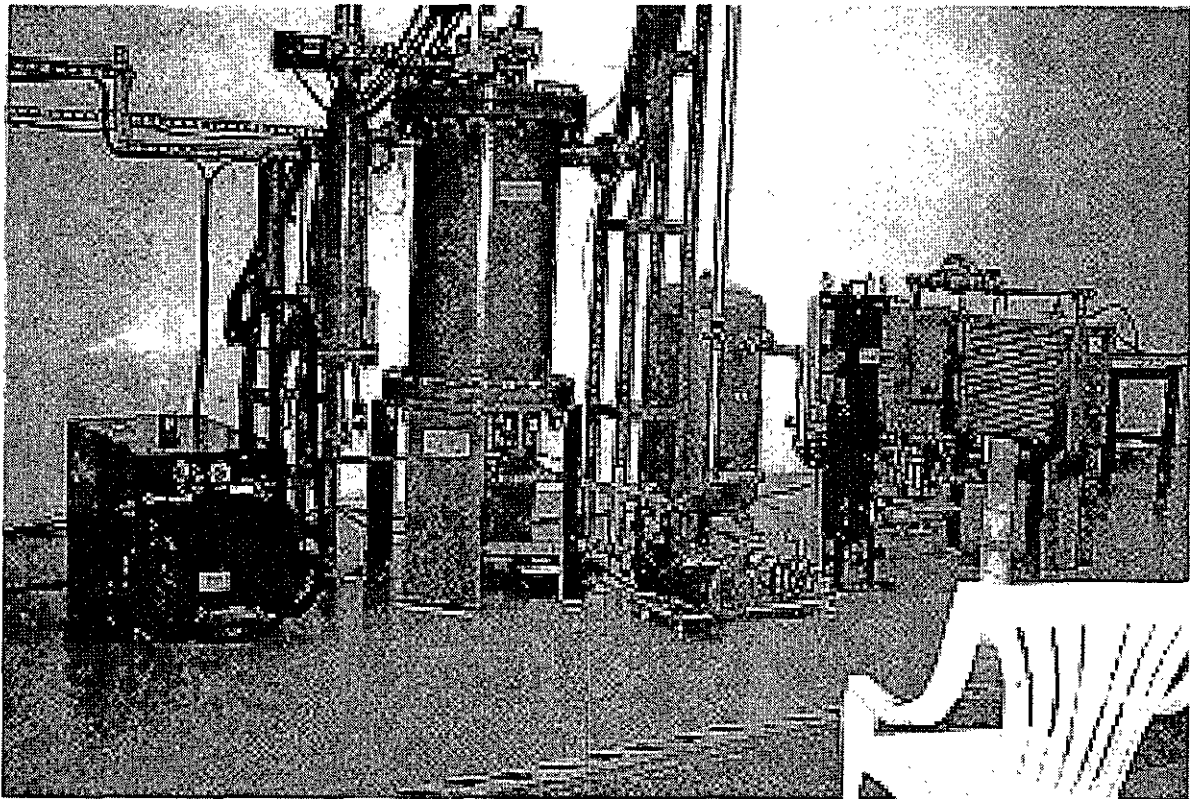
Once the self-sufficient production of clean energy is established via measurements for Hadronic Power Plants of the First Type, investments for the developments of the corresponding power plants of the Second and Third Type have minimal risk (of course, under sufficient development funds) since catalytic liquefaction and nuclear syntheses merely increase the self-sufficiency and the energy output.

2.2: Efficiency of the PlasmaArcFlow Process

The "Scientific Energy Efficiency" (SEE) of the PlasmaArcFlow Reactors is defined by

$$(2) \text{ SEE} = \text{Energy Out} / \text{Energy In} < 1,$$

and it is always smaller than one because of the conservation of the energy and the unavoidable losses. The energy out is given by heat and MagneGas, while the energy in is given by the electric energy used by the arc plus the energy in carbon. We, therefore, have:



$$(3) \text{ SEE} = [\text{Heat Energy} + \text{MagneGas Energy}] / [\text{Electr. Energy} + \text{Carbon Energy}] < 1.$$

The electrodes are made of coal and not graphite, because graphite is needed for accurate welding while the use of coal in the reactors produces MagneGas with bigger energy content. Also, coal is quite inexpensive and its cost is generally compensated for by the income from the recycling of liquid wastes. As such, carbon is generally ignored in the industrial applications of the technology. Also, as indicated earlier, for a submerged electric arc we have

(4) Carbon Energy >> Electr. Energy.

The above aspects imply the following "Commercial Energy Efficiency" (CEE):

(5) $CEE = [\text{Energy in Heat} + \text{Energy in MagneGas}] / [\text{Electr. Energy}] \gg 1.$

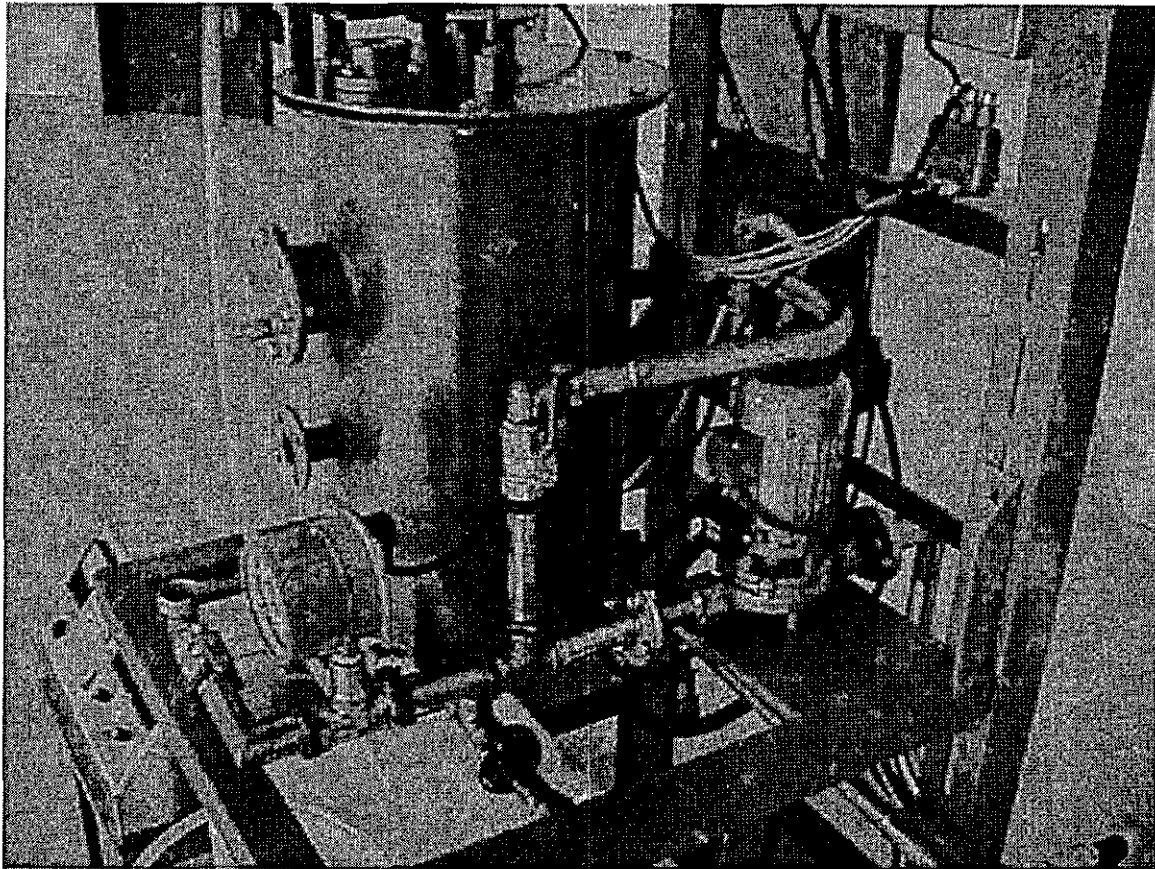


FIGURE: A view of the research PlasmaArcFlow Recycler in Kazakhstan

It should be stressed that energy cannot be created from nothing, as it is hoped for the "zero point energy" and other attempts. Therefore, the above "over-unit" is purely "commercial" and it is merely due to the cheap cost of coal permitting its removal in the denominator of the scientific energy efficiency (3). The definition of the energy efficiency of hadronic reactors requires numerous additional terms, but they are too technical for this introductory presentation.

It is important to know that the above commercial energy efficiency is not constant and it can be increased as follows:

1) **The Commercial Energy Efficiency increases with the increase of the operating pressure.** This is due to the fact that at atmospheric pressure, the electric arc mostly occurs within the produced gas, and only a small part occurs within the liquid. With the increase of the pressure the bubbles of gas surrounding the arc decrease, with

consequential increase of the travel of the arc through the liquid. It then follows that the increase of pressure implies an increase of the Commercial Energy Efficiency. This increase has been established by measurements, although the general dependence has not been plotted to date, and the efficiency at very high pressures is vastly unknown. Note that the increase is nonlinear in the sense that the doubling of the pressure does not imply the doubling of the efficiency but only its increase of the order of 5%-10% depending on other aspects.

2) The Commercial Energy Efficiency increases with the increase of the power. This is due to the fact that the volume of gas produced is directly proportional to the gap between the electrodes, the longer the gap, the bigger the volume of produced gas. In air such a gap can be large, but for a submerged electric arc the only possibility of increasing the arc gap is via the increase of the power, since the liquid medium is an insulator. Again, the increase is nonlinear in the sense that the doubling of the power does not imply the doubling of the efficiency, but certainly implies increases of another 5% to 10%.

3) The Commercial Energy Efficiency increases with the increase of the operating temperature. This is due to the fact that the greatest portion of the electric energy is used for the evaporation of the water. It is then evident that the higher the temperature, the bigger is the amount of electric energy used for water separation (rather than water evaporation), resulting in an increase of efficiency that is again nonlinear in the sense indicated above.

4) The Commercial Energy Efficiency generally increases with the increase of the carbon content of the selected liquid feedstock, and it is in fact maximal for oil-base liquids.

5) The Commercial Energy Efficiency also increases under the appropriate selection of additives intended to increase the thermal energy release of the chemical reactions in the arc plasma.

The Commercial Energy Efficiency of hadronic recyclers also increases with other factors, the biggest increases being permitted by the new arc processes under development at this writing we cannot disclose for security reasons.

In conclusion, no appraisal of the energy efficiency of hadronic reactors is credible or otherwise meaningful, without a consideration of the operating pressure, power, temperature, selected liquid feedstock, selected additive and other factors. Note that the electric arc is insensitive to operating pressures up to very high values to the order of 100,000 psi (over 6,000 bars) that are not attainable today. Therefore, PlasmaArcFlow can be constructed up to the currently allowed 10,000 psi (666 bars).

The increase of power merely requires suitable re-design. In fact, the technology of arc furnaces is well developed to use 1 MW electric arcs between electrodes of up to 3' x 8' (90 cm x 240 cm). Such a technology with very high powers only needs to be adapted to

hadronic recyclers, of course, depending on available funds. Finally, the operating temperature is only due to the amount of heat extracted via heat exchangers, the ideal operating conditions being evidently those in which the selected operating temperature is kept constant with heat exchangers by the sole removal of all excess heat produced.

The ideal industrial recycler is one that has at least 1 MW, operates at about 5,000 psi (333 bars) and at a temperature higher than the boiling temperature of the selected liquid at 5,000 psi (333 bars).

At this moment we have available the 50 KWh reactor in Verona operating at atmospheric pressure and temperatures, and the 100 KWH industrial recycler available in Florida operating up to 150 psi (10 bars) and about 300 degrees F, plus a research recycler available in Kazakhstan. Despite their limitations, these reactors can indeed establish the self-sufficient generation of clean energy with Hadronic Power Plants of only the First Type.

A serious commitment to our serious environmental problems requires INVESTMENTS as needed to measure the efficiency at various conditions, and then to build the prototype Hadronic Power Plant of the First Type. The extension of such a plant to those of second and third type will only increase the clean energy produced.

2.3: Commercial Efficiency of the Verona Reactor

The recycler currently available in Verona has 50 KWh, operates at a maximum of 7 psi (0.5 bar) and has no heat exchanger to maintain the heat constant. Also, the reactor has no means to measure the efficiency, since it lacks a cumulative wattmeter to measure the energy used per each test, and a flowmeter to measure the volume of MagneGas produced. As a result, all data below are approximated and essentially based on measurements conducted in the U.S.A. in similar recyclers.

Under the above clarifications, it is expected that in the Verona Reactor the use of 100 W-h produces 1.5 scf-h of MagneGas (42 L-h), plus about 300 BTU-h of heat. By recalling that $1 \text{ W} = 3.14 \text{ BTU}$ and that MagneGas produced from water has the energy content of about $700 \text{ BTU/scf} = 25 \text{ BTU/L}$ (depending on its combustion - see below), we have the following predicted Commercial Energy Efficiency of the Verona Reactor

$$(6) \text{ CEE} = [300 \text{ BTU of heat} + 1,050 \text{ BTU of MagneGas}] / [314 \text{ BTU Electr.}] = 4.29 > 1.$$

Despite the fact that the Verona Recycler was built for a simple illustration of the technology and absolutely not to reach self-sufficiency production of electricity, it is easy to see that the recycler as currently available has indeed a positive energy output after the self-production of the needed electricity.

The 50 KWh Miller Dimension 1000 used by the recycler has an efficiency of about 90% as per manufacturer's specs. This implies that the arc actually uses $90 \text{ W-h} = 280 \text{ BTU-h}$ of DC electricity per each 1.5 scf (42 L-h) of MagneGas produced. It is then easy to see that the energy content of MagneGas is expected to be sufficient for the production of

usable energy. In fact, the use of a conventional DC (rather than AC) generator based on internal combustion engines with an efficiency of 38% characterizes the following:

(7) Self-Generation of Electricity: $38\% \ 1,050 \text{ BTU} = 399 \text{ BTU} > 280 \text{ BTU}$ of the arc,

Therefore, the Verona Recycler, as available now, is expected to have the following usable heat generation after the self-production of DC electricity

(8) Usable Heat Generation: $(119 \text{ BTU-h} + 300 \text{ BTU-h}) \times 50 = 20,950 \text{ BTU-h}$

By assuming that also a 38% conversion into usable electricity, the Verona Reactor is predicted to have the following self-sufficient production of usable electric power

(9) Usable Power Generation: $38\% \ 20,950 \text{ BTU-h} = 7,961 \text{ BTU-h} = 2.53 \text{ KWh}$

It should be noted that, in the current state of construction, the efficiency of the Verona Recycler cannot be increased with the increase of pressure beyond 7 psi (0.5 bar) because of the inability to seal the pressure around carbon rods. Nevertheless, without any modification, the efficiency can be increased via the recycling of a mixture of oil-base liquids and water.

In this case 100 W-h of AC electricity or 90 W-h of DC electricity are expected to produce 2 scf-h (56 L-h) of MagneGas with about 1,000 BTU/scf (35 BTU/L) and about 500 BTU of heat, resulting in the predicted Commercial Energy Efficiency

(10) $\text{CEE} = [500 \text{ BTU heat} + 2,000 \text{ BTU MagneGas}] / [314 \text{ BTU Electr.}] = 7.96 \gg 1$

with the following percentage use of MagneGas for the self-generation of DC electricity at 38% efficiency

(11) $38\% \ 736 \text{ BTU-h} = 280 \text{ BTU-h}$ of the electric arc,

thus leaving the following usable power

(12) $500 \text{ BTU-h heat} + 1,264 \text{ BTU-h MagneGas}] \times 50 = 88,200 \text{ BTU-h}$

that, again converted at 38% efficiency, permits the prediction of the following usable power:

(13) Usable Power Generation: $38\% \ 88,200 \text{ BTU-h} / 3,14 = 10.67 \text{ KWh}$.

It should be noted that, under the availability of the necessary funds, the Verona Reactor can be easily modified into a Hadronic Power Plant of the First Type via: 1) The replacement of the current low pressure PlasmaArcFlow Module with a high pressure one; 2) The addition of a heat exchanger for the production of steam, and 3) The addition of a DC generator operated via a turbine powered by steam as well as MagneGas.

This modification will allow the entire use of the produced heat plus the partial use of MagneGas produced for the self-generation of electricity. The balance of available MagneGas can be used for the production of electricity or other uses.

Independent efficiency measurements should also be conducted with the recycler in Florida. Unfortunately, the geometry of the PlasmaArcFlow in the latter Reactor is open, as compared to that with the Venturi of the Verona Recycler. As a result, the Florida Reactor should not be used for comparison with the Verona Recycler, and should be used instead for the purposes for which it was built, namely, establish the dependence of the efficiency with the increase of power, pressure and temperature, to verify the existence of the Controlled Hadronic Fusion, and then to complete the construction into a fully operational prototype Hadronic Power Plant.

2.4: Energy Efficiency of the Hadronic Power Plants of Second Type

In the Hadronic Power Plants of the First Type the primary source of energy for the self-generation of the electricity is the MagneGas fuel. For the case of plants of the second type, the primary source of energy for the DC arc is expected to be the heat produced by the arc itself plus that produced by the catalytic liquefaction of MagneGas, which heat is utilized via a heat exchanger to produce steam, plus a small percentage of MagneGas produced to bring steam to supercritical temperatures for a more efficient turbine operation (numerous other means are possible in lieu of burning MagneGas, such as solar, wind, or chemical sources of additional energy).

This case is treated in detail in the US patent [4] and it is not repeated here for brevity. We only note that, as it is well known, heat is the most difficult form of energy to be utilized due to its large natural dissipation through walls. In US Patent [4] a large heat loss has been assumed due to dissipation. Despite that, sufficient heat for the self-generation of electricity is indeed achieved with excess energy available in the form of Liquid MagneGas.

2.5: Efficiency of Hadronic Power Plants of Third Type

As recalled in Section 1, despite huge public investments, "hot fusion" has now been demonstrated to have no foreseeable industrial value. The primary reason for this failure is the excessive temperature under which instabilities caused by the activation of the fusion cannot be controlled.

We have also indicated in Section 1 that the "cold fusion" has indeed established the capability of producing nuclear fusion in our laboratories, although without any foreseeable industrial significance because the energy gains are quite marginal, as well known. The reason for this occurrence is opposite that of the "hot fusion," and it is due to insufficient temperatures at which nuclei remain generally shielded by their electron clouds, in which case no systematic nuclear synthesis can possibly occur with an industrial relevance.

In view of these two shortcomings, the author has developed the new "Controlled Hadronic Fusion" whose main principle is that of operating at intermediate temperatures between those of the cold and hot fusions, so as to allow nuclei to be free of their electron clouds as a necessary pre-requisite for their fusion, although without excess temperatures causing uncontrollable instabilities.

The new "Controlled Hadronic Fusion" is also based on other principles, the most important one being the true control of ALL aspects necessary for the desired fusion, including the control of the temperature, pressure, power, and other data.

The details of the new Controlled Hadronic Fusion cannot be disclosed here for security reasons. We merely mention that the electric arc constitutes one of the best means for its realization because it is the ideal mean for controlling ionization, pressure and temperature.

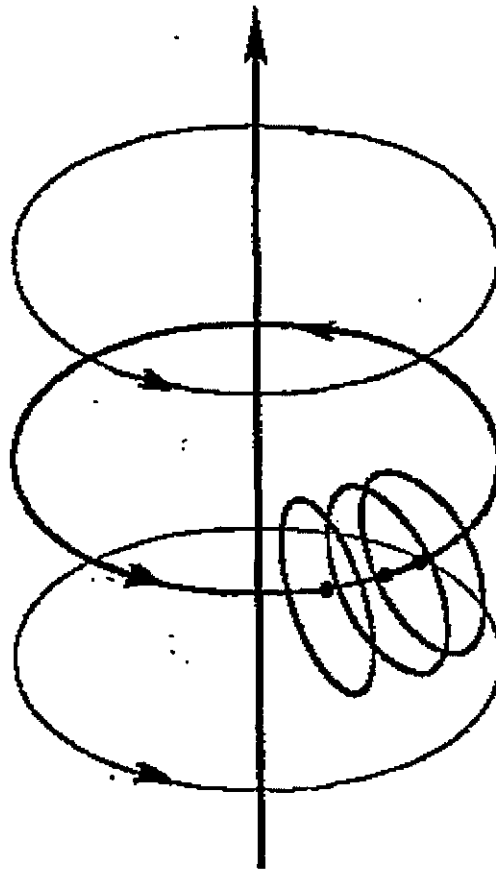
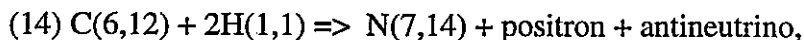


FIGURE: A schematic view of the principles of the Controlled Hadronic Fusion. Under the proper control of power, pressure and temperature, the electric arc permits: 1) The polarization of electron orbitals along toroids perpendicular to the local magnetic force field as the basic condition to expose nuclei at moderate temperatures; 2) The alignment of polarized atoms along opposite magnetic polarities North-South-North; and 3) The compression of polarized and aligned atoms due to Coriolis and other effects.

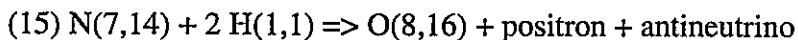
The evidence of the existence of the Controlled Hadronic Fusion is the following: As recalled earlier, the biggest source of heat in the processing of water with PlasmaArcFlow Reactors is the creation of CO with 255 Kcal/mole. It is very easy to verify that, by far, the largest amount of heat is produced when using crude oil or engine oil waste as liquid feedstock. In fact, the heat produced is such that, with insufficient flow of the liquid feedstock, even the carbon electrodes would melt, thus implying the achievement of local temperatures near the arc of the order of 50,000-70,000 degrees F.

Nuclear synthesis is the sole credible origin of the large amount of heat produced when using oil-base liquid feedstock. This is due to the fact that, having the generic chemical composition C_nH_{2n} , oil contains no oxygen at all. The electrodes are composed of coal, and they too have no oxygen at all. Similarly, all reactors are primed with helium (or other inert gases) before the initiation of any operation for safety reasons. Therefore, when operated with oil base liquid feedstock, hadronic reactors have no oxygen at all.

Lacking the oxygen, there exists no known chemical reaction capable of interpreting quantitatively the very large amount of heat produced, thus leaving nuclear synthesis as the only credible explanation. Numerous syntheses are possible but we cannot outline them here for security reasons. We merely mention that the first synthesis preferred by nature is precisely that of the nitrogen, Eq. (1), i.e.



The next synthesis preferred by nature is that of the oxygen synthesized by the nitrogen plus the large number of hydrogen atoms available in the arc's surroundings,



In fact, beginning with a clean reactor, water is routinely found at the bottom of the oil-base liquid feedstock in macroscopic amounts that cannot be explained via humidity and similar marginal sources.

Once confirmed, this evidence is expected to be sufficient for investors with a vision. Industrial equipment will merely depend on the invested funds.

SECTION 3: PROPOSED TESTS AND THEIR PROJECTED COSTS

3.1: Introduction

As a scientist, the author cannot participate in any additional study of Hadronic Power Plants without actual measurements. In turn, measurements require suitable meters specifically selected for the needed data. The prototype Hadronic Power Plant from the Florida recycler is then recommended for construction. After measurements on the basic data, laws and behavior are available, the construction of an industrial Hadronic Power Plant is suggested. In the following we identify the needed measurements, related detectors and projected costs.

3.2: Measurement of Efficiency of the Verona Recycler

To achieve any meaningful information, we need to do the following:

1) Measuring the electric power used from the grid. In the U.S.A. we have routinely done this measurement with an independently calibrated, cumulative Wattmeter with digital display, so as to have a reliable reading of the total watt-hour used per test.

2) Jointly, we need an accurate measurement of the volume of MagneGas produced. This requires a very special flowmeter that is not affected by the magnetic anomalies of the gas, and, to have any value, the flowmeter needs to be calibrated with a rather complex procedure requiring the combustion of a special apparatus, such as a tower of known volume filled up with water that is displaced by the MagneGas produced so as to achieve a valid measurement of volumes of MagneGas produced from which the flowmeter can then be calibrated.

3) The third basic need is to reach some reliable value of the energy content of MagneGas. At this point most engineers and scientists with zero knowledge of the MagneGas Technology suggest the use of conventional calorimeters. This trend is based on the old knowledge that all conventional fuels have a fixed thermal content irrespective of their uses. By comparison, the anomalous character of MagneGas is established precisely by the fact of having variable energy content depending on its use. In particular, no calorimeter has provided any valuable information on MagneGas. For instance, calorimeters produced for natural gas generally do not even allow MagneGas to burn, let alone measure anything, due to the large differences in stoichiometric ratios between natural gas and MagneGas. Bomb calorimeters are only useful to provide false "experimental beliefs" for several reasons, such as the fact that they do not allow the measurement of the exhaust to make sure that all fuel did indeed burn before venturing claims of experimental results on heat content. A design of a calorimeter specially intended to measure the energy content of MagneGas is available, but its construction cost is estimated in excess of \$100,000 due to the numerous functions requested for a serious measurement, such as variable stoichiometric ratios, variable ignition means, measurements at variable pressures, etc. In view of these aspects, the test here suggested is the one used until now in the U.S.A., namely, the measurement of the comparative energy content of MagneGas and natural gas in specific equipment, such as an electric generator, or the measure of power of a car on a dynamometer. It should be indicated that, in view of a long history of expensive failures in calorimeter measurements imposed on the author in the past by associates in the U.S.A., the author will participate in no measurement of the energy content of MagneGas via conventionally available calorimeters, and will terminate any collaboration in case such measurements are done in view of their evident major damage to the MagneGas Technology. The reason is that calorimeter measurements produce values of energy content of MagneGas way below realities, which low values may be accepted and propagated for political objectives against the MagneGas Technology.

4) Next, we have to measure the heat produced by the arc. This can be achieved via simple measurements of temperatures before and after each test (with due lapsed times

for thermal stability). Knowledge of the thermal coefficient of the used liquid feedstock and of the steel of the reactors as well as their masses produces reliable results.

5) None of the above measurements has any value without measuring the combustion exhaust of MagneGas, which requires a permanently available 4-ways analyzer (for HC, CO, CO₂ and O₂) with computer print-out. As an illustration, the author knows how to produce MagneGas with 2,000 BTU/scf, that is, double the thermal content of natural gas, under which we have a large self-sufficient production of usable power. However, this solution is useless because the flame of MagneGas with 2,000 BTU/scf is black, thus being not recommendable for any practical use due to excessive pollution.

The costs of the above measurements in U.S. dollars (and based on U.S. suppliers) is projected to be the following:

Cumulative Wattmeter \$600
Independent calibration of same \$1,000
MagneGas Flowmeter \$7,000
Independent calibration of the same \$5,000
10 KWh Natural Gas Electric Generator \$3,000
Optimization for MagneGas by combustion experts \$3,000
Thermometers \$300
4-ways analyzer \$7,000
Chemical analyses \$3,000
Personnel, overheads and misc . \$10,000
Miscellaneous, unpredictable events, reserves \$10,000

TOTAL PROJECTED COST \$49,900

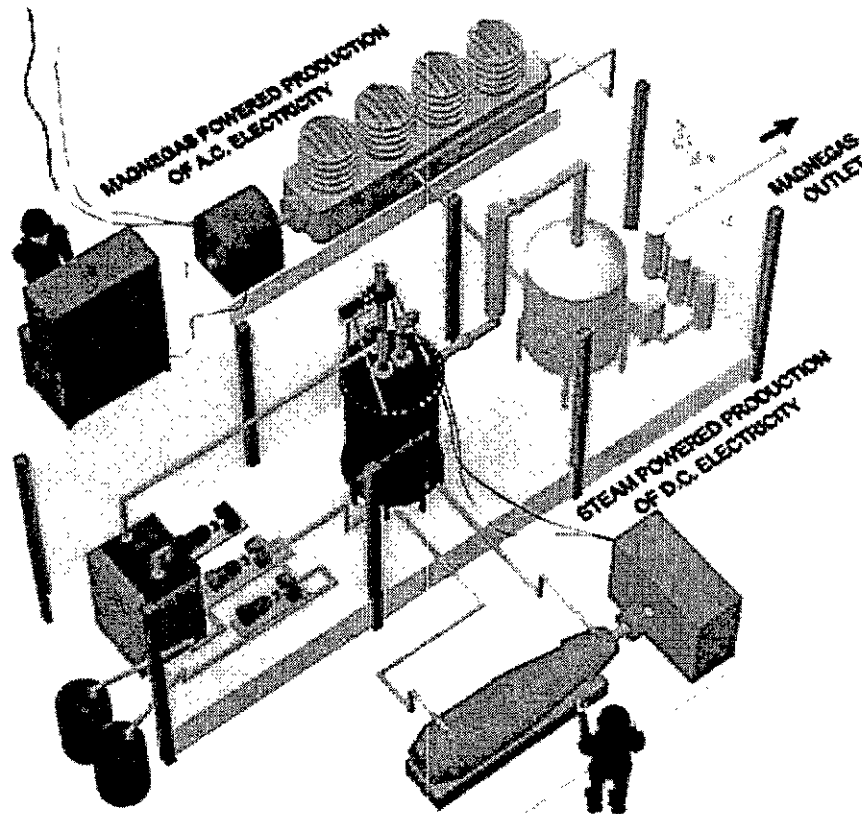
3.3: Measurement of Efficiency of the Florida Reactor

As indicated earlier, the data on the Florida Reactor cannot be compared to those of the Verona Reactor because of a large difference in the geometry of the PlasmaArcFlow (the Verona Reactor being better because of a Venturi surrounding the arc).

The Florida Reactor was built for the different scope of first measuring and plotting the values of the produced heat and MagneGas as a function of the pressure, power and temperature. Once these data were measured and verified, the Reactor was built to be completed into the prototype Hadronic Power Station of the first type. The primary reason for the expensive construction of the industrial reactor in Florida at high pressure has been the verification of the new Controlled Hadronic Fusion.

The needed measurements are rather complex, each of them requiring considerable time, each measurement also requiring its conduction with different liquid feedstocks (for instance, comparative values of heat production in water-base and oil-base liquid feedstocks). As such, the needed measurements are expensive in cost and time.

Firstly, the Florida Reactor must be completed with two additional 50 KWH power units to permit meaningful plotting of the data as a function of the power from 30 KWH to 150 KWH. Secondly, the recycler has to be completed with the same Wattmeter and flowmeter needed for the Verona Reactor and various other sensors and detectors. Thirdly, the reactor has to be equipped with a remote control for the necessary safety when operating at high pressure. These initial costs are projected to be to the order of \$50,000.



DIFURE: A 3D view of the proposed prototype Hadronic Power Plant built via the use of the 100 KWh PlasmaArcFlow Recycler currently available in Florida.

Also, new facilities need to be rented in a suitably isolated area for quiet conduction of all needed tests at high pressure without danger to personnel or neighbors. The cost of the rental, the contract with the electric company for 150 KWh power, and the construction of the necessary related electric panel are expected to exceed \$50,000.

Technicians have to be hired and trained for extensive tests and numerous outside companies have to be used for independent verifications estimated to cost at least \$50K.

At that moment the most important measurement to be initiated are those regarding the Controlled Hadronic Fusions, which cannot be identified here for security.

Following all the above, the Florida Recycler should be completed into a prototype 100 KWH Hadronic Power Plant with the addition of: 1) A 100 KWh electric generator produced for natural gas and used with MagneGas; 2) a heat exchanger producing steam with a separate station bringing steam to supercritical temperature; and 3) A DC generator operated by a turbine powered by steam and MagneGas.

Detailed drawings of the completion of the Florida Recycler into a prototype Hadronic Power Plant are already available, including a 3D view. The operations are initiated with a cold liquid via the activation of the turbine powered DC generator with the sole combustion of MagneGas from preceding runs. The combustion of MagneGas is decreased with the initiation of production of steam in about 5 minutes, and finally reduced to a minimum when full operational temperature is reached in about 10 minutes.

Two solutions are possible for the implementation of the above program. First, the Florida Recycler can be shipped to Verona and all tests done jointly for the two recyclers, including the construction of the prototype Hadronic Power Plant. The author has contacted the U.S. investors who have financed the construction of the high pressure Florida reactor and they have agreed to allow its shipment to Italy (together with all other equipment available) upon reimbursement of its cost of \$250,000 (which includes no payment to the author for his work at that recycler). Notice that, according to long standing practice, no shipment of equipment is possible from the U.S.A. unless paid in full before leaving U. S. soil. Notice that, according to this first solution, in addition to the funds needed for the purchaser of the Florida Recycler, all tests would have to be funded in Italy by the Compagnia di Verona.

The second alternative is that of conducting all tests of the Florida Reactor in the U.S.A. In this case the best solution for the funding of the cost is the investment of \$500,000 in a significant equity position of USMagneFuels, Inc., the Florida corporation with the exclusive rights of the MagneGas Technology for the American Continent (only), under the proviso that all funds shall be devoted to the indicated tests, and all results shall be available to the Compagnia di Verona. It should also be indicated that, since USMagneFuels, Inc., is currently seeking financing, investments for more than \$500,000 are possible, all the way to the acquisition of control depending on the volume of funds invested. Evidently, the best solution would be the progressive investment of funds, beginning with an initial investment, and then increasing the investment depending on results. Detailed financial forecasts of USMagneFuels, Inc. are available on request to seriously interested investors.

The author recommends the second solution for various reasons, such as the evident need for the developments of MagneGas Technology in the European and American continents to be deeply integrated and correlated; the fact that the author will remain a Florida resident and can only visit Italy on a periodic basis; and the availability in Florida of trained experts.

3.4: Liquefaction of MagneGas

A Fisher-Trophy tower is recommended to be built for testing the liquefaction of MagneGas, and selecting the needed catalysts and operating pressure. The combustion exhaust of the liquid fuel so obtained must be individually tested because of the possible need for different additives in the MagneGas production (see for details patent [4]).

The costs for the prototype production of Liquid MagneGas are estimated to be to the order of \$100,000 including the construction of a Fisher Trophy station that can be added to existing recyclers.

It should be noted that the industrial value of the results are directly proportional to the investment in R&D. The first objective is evidently the achievement of a liquid MagneGas with clean combustion exhaust. However, under the availability of additional R&D funds it is possible to seek a liquid MagneGas with very low vapor pressure, in which case the new liquid fuel can be used in gasoline tanks of available cars.

The reader should appraise the immense commercial implications in achieving a Liquid MagneGas with low vapor pressure suitable for use in existing gasoline tanks, and the large lack of gains that may originate from insufficient R&D funding in such a project.

3.5: First Industrial Hadronic Power Plant

The projected total cost for the design, construction and extensive test of the first industrial Hadronic Power Plant of 1MWh power, from its initiation to the completion of all tests, including revised drawings ready for production, is expected to be \$1M.

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